

Extending not-MIWAE for correlated missing not at random mechanisms

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1 Introduction

Missing data are common in real-world applications, from healthcare and social sciences to recommender systems and sensor networks, and their patterns of missingness are often complex, especially when missingness depends on the missing values. For example, in social science surveys, people at extreme income levels are less likely to answer income-related questions. If not accounted for, such patterns can bias machine learning models and reduce predictive accuracy. How we handle incomplete data depends on why the values are missing, i.e., the dependencies of the missing mechanism. There are three main categories of missing mechanisms:

- **Missing Completely At Random (MCAR):** Missingness is independent of data.
- **Missing At Random (MAR):** Missingness depends only on observed values.
- **Missing Not At Random (MNAR):** Missingness may depend on both observed and unobserved values.

For MNAR imputation, the not-MIWAE model (Ipsen, Mattei, and Frellsen 2020) represents a major advance in jointly modeling data and missingness, but it assumes that missingness is conditionally independent given the data, a condition often violated in practice. In this project, we extend not-MIWAE by introducing a new latent variable pointing directly at the mask in order to capture dependencies among mask variables. Experiments on five UCI datasets show that our model outperforms not-MIWAE in imputation accuracy across all datasets.

2 The not-MIWAE model

This model is a direct extension of MIWAE (Mattei and Frellsen 2019) for MNAR scenarios, which in turn is built upon the importance weighted autoencoder (IWAE) framework proposed by Burda, Grosse, and Salakhutdinov 2015. In this section, we introduce the mathematical formulation of not-MIWAE.

^{*}Contribution statement: Both authors contributed to all aspects of the project. Mathéo derived the key theoretical results of the extended not-MIWAE model and adapted the authors' code for the extension. Yee-Yang was responsible for the experimental results.

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Notations. Consider a dataset of n independent observations $X = (x_1, \dots, x_n)^T \in \mathcal{X}^n$, where each $x_i \in \mathcal{X}$ lies in a p -dimensional feature space $\mathcal{X} = \mathcal{X}_1 \times \dots \times \mathcal{X}_p$. In the presence of missing values, each observation splits into observed and missing parts: $x_i = (x_i^o, x_i^m)$, determined by a binary indicator vector $s_i \in \{0, 1\}^p$ where $s_{ij} = 1$ indicates that feature j is observed in sample i , and $s_{ij} = 0$ indicates it is missing. We collect all indicators in the mask matrix $S = (s_1, \dots, s_n)^T \in \{0, 1\}^{n \times p}$.

The objective is to model the joint distribution $p_{\theta, \phi}(x, s) = p_{\theta}(x)p_{\phi}(s|x)$. We seek to maximize the log-joint likelihood of observed quantities:

$$\ell(\theta, \phi) = \sum_{i=1}^n \log p_{\theta, \phi}(x_i^o, s_i) \quad (1)$$

The model architecture (Figure 1) consists of two components: a latent variable model $z \sim p(z)$ generating data via $p_{\theta}(x|z)$, and a missing mechanism $p_{\phi}(s|x)$ predicting masks from complete data.

Since $p_{\theta}(x|z) = \prod_j p_{\theta}(x_j|z)$ factorizes, each term becomes:

$$\log p_{\theta, \phi}(x^o, s) = \log \int \int p_{\phi}(s|x^o, x^m) p_{\theta}(x^o|z) p_{\theta}(x^m|z) p(z) dz dx^m$$

(2) Figure 1: Graphical model of not-MIWAE

This double integral is computationally intractable. To address this, we introduce an inference network $q_{\gamma}(z|x^o)$ as an importance sampling proposal, transforming the expression into:

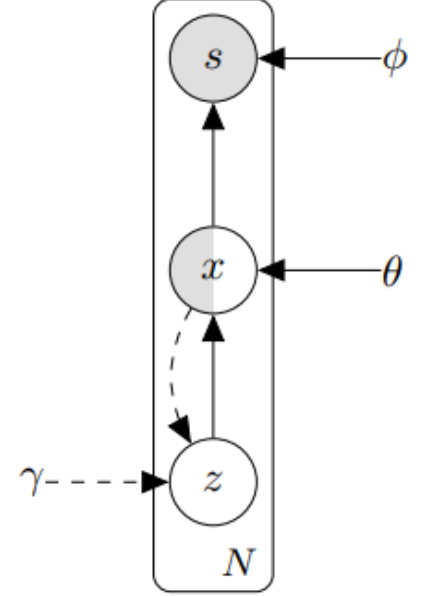
$$\log p_{\theta, \phi}(x^o, s) = \log \mathbb{E}_{z \sim q_{\gamma}(z|x^o), x^m \sim p_{\theta}(x^m|z)} \left[\frac{p_{\phi}(s|x^o, x^m) p_{\theta}(x^o|z) p(z)}{q_{\gamma}(z|x^o)} \right] \quad (3)$$

Following the IWAE framework Burda, Grosse, and Salakhutdinov 2015, we approximate this expectation using K Monte Carlo samples:

$$\mathcal{L}_K(\theta, \phi, \gamma) = \sum_{i=1}^n \mathbb{E} \left[\log \frac{1}{K} \sum_{k=1}^K w_{ki} \right], \quad w_{ki} = \frac{p_{\phi}(s_i|x_i^o, x_{ki}^m) p_{\theta}(x_i^o|z_{ki}) p(z_{ki})}{q_{\gamma}(z_{ki}|x_i^o)} \quad (4)$$

where samples (z_{ki}, x_{ki}^m) are drawn from $q_{\gamma}(z|x_i^o) p_{\theta}(x^m|z)$. Gradients flow through both sampling steps via reparameterization.

The missing mechanism factorizes as $p_{\phi}(s|x) = \prod_{j=1}^p \text{Bern}(s_j|\pi_{\phi,j}(x))$, with various architectural choices for computing $\pi_{\phi,j}(x)$ (feature-specific, global, or constrained). Our extension relaxes this independence assumption.



3 Extended not-MIWAE model

As seen in equation (10) of Ipsen, Mattei, and Frellsen 2020, the mask variables S_j are assumed to be conditionally independent given input x :

$$p_\phi(s \mid x_o, x_m) = p_\phi(s \mid x) = \text{Bern}(s \mid \pi_\phi(x)) = \prod_{j=1}^p \pi_{\phi,j}(x)^{s_j} (1 - \pi_{\phi,j}(x))^{1-s_j}. \quad (\text{not-MIWAE eqn. 10})$$

This assumption is theoretically appealing but may not hold in practice, where missingness in one column can affect that of another column (e.g. skipping a medical test that results in three missing metrics, or survey questions that only require an answer if the respondent answers “yes” to the previous question). We relax this assumption in several ways that we consider realistic.

We introduce an additional latent variable $u \sim \mathcal{N}(0, I)$ representing external factors influencing missingness, independent of data generation. The joint distribution becomes:

$$p_{\theta,\phi}(x, s, z, u) = p_\phi(s|x, u) p_\theta(x|z) p(z) p(u) \quad (5)$$

The missing predictor now conditions on both data and regime:

$$p_\phi(s|x, u) = \prod_{j=1}^p \text{Bern}(s_j | \pi_{\phi,j}(x, u)) \quad (6)$$

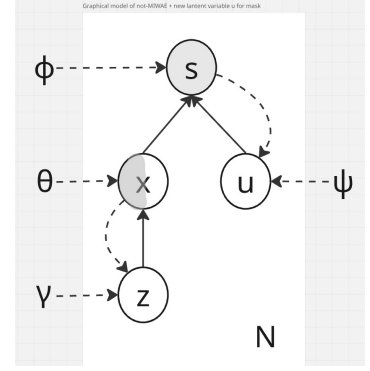


Figure 2: Extended graphical model with regime variable u

While factorization over features j is maintained, the shared latent u captures dependencies between mask variables (e.g., collection protocols affecting multiple features, measurement regimes, latent confounders).

For inference over u , we introduce a regime encoder. Two approaches are relevant:

$$q_\psi(u|x^o, s) = \mathcal{N}(u | \mu_\psi(x^o, s), \sigma_\psi(x^o, s)) \quad (\text{full encoder : used in the majority of the work}) \quad (7)$$

$$q_\psi(u|s) = \mathcal{N}(u | \mu_\psi(s), \sigma_\psi(s)) \quad (\text{mask-only encoder}) \quad (8)$$

The complete inference uses three proposals: $z \sim q_\gamma(z|x^o)$, $x^m \sim p_\theta(x^m|z)$, and $u \sim q_\psi(u|x^o, s)$ (or $q_\psi(u|s)$).

The joint likelihood becomes:

$$\log p_{\theta,\phi}(x^o, s) = \log \iiint p_\phi(s|x^o, x^m, u) p_\theta(x^o|z) p_\theta(x^m|z) p(z) p(u) dz dx^m du \quad (9)$$

Introducing both variational distributions $q_\gamma(z|x^o)$ and $q_\psi(u|x^o, s)$ as importance sampling proposals:

$$\log p_{\theta,\phi}(x^o, s) = \log \mathbb{E}_{\substack{z \sim q_\gamma(z|x^o) \\ u \sim q_\psi(u|x^o, s) \\ x^m \sim p_\theta(x^m|z)}} \left[\frac{p_\phi(s|x^o, x^m, u) p_\theta(x^o|z) p(z) p(u)}{q_\gamma(z|x^o) q_\psi(u|x^o, s)} \right] \quad (10)$$

Following the same importance sampling procedure as in standard not-MIWAE, we obtain the extended lower bound:

$$\mathcal{L}_K(\theta, \phi, \gamma, \psi) = \sum_{i=1}^n \mathbb{E} \left[\log \frac{1}{K} \sum_{k=1}^K w_{ki} \right] \quad (11)$$

with importance weights:

$$w_{ki} = \frac{p_\phi(s_i | x_i^o, x_{ki}^m, u_{ki}) p_\theta(x_i^o | z_{ki}) p(z_{ki}) p(u_{ki})}{q_\gamma(z_{ki} | x_i^o) q_\psi(u_{ki} | x_i^o, s_i)} \quad (12)$$

The network $\pi_{\phi,j}(x, u)$ can be implemented via concatenation, additive, or multiplicative architectures, each capturing different interaction patterns between data and regime variables:

- **Concatenation:** $\pi([x, u]) = \sigma(\text{MLP}([x, u]))$
- **Additive:** $\pi(x, u) = \sigma(\text{MLP}_x(x) + \text{MLP}_u(u))$
- **Multiplicative:** $\pi(x, u) = \sigma(\text{MLP}_x(x) \times \text{MLP}_u(u))$

4 Experiments

Ipsen, Mattei, and Frellsen 2020 shows that not-MIWAE outperforms competing models in the literature, such as missForest (Stekhoven and Bühlmann 2012) and MICE (Van Buuren and Groothuis-Oudshoorn 2011), in terms of imputation RMSE on datasets affected by MNAR. In this section, we introduce a more complex MNAR mechanism that relaxes conditional independence between mask variables (not-MIWAE eqn. 10). Table 1 shows that our extended model with the full encoder (Equation 7) achieves better imputation results than not-MIWAE in addition to several selected models across all five datasets¹.

Regarding architecture choice of the network $\pi_{\phi,j}(x, u)$, for all datasets except *Breast*, the extended not-MIWAE model with the concatenation architecture for the missingness predictor $\pi_{\phi,j}(x, u)$ achieves the lowest imputation RMSE among eight selected models, with the baseline not-MIWAE model using the nonlinear missingness architecture as a close second. For the *Breast* dataset, the extended not-MIWAE model achieves the best performance with the other two architectures, additive and multiplicative.

4.1 Setup

Data. We train the models on five of the six UCI datasets used in Ipsen, Mattei, and Frellsen 2020. Since the authors’ publicly available source code only includes the replication package for the *White* dataset, the results may differ slightly, as the remaining datasets may have been prepared differently or trained with different parameter settings. The *Yeast* dataset was excluded because we

1. Performance degrades when using the mask-only encoder (Equation 8) Results can be found in the Appendix (Table 3).

Model	Banknote	Concrete	Red	White	Breast
MIWAE	1.46 ± 0.00	1.47 ± 0.01	1.44 ± 0.05	1.40 ± 0.08	1.20 ± 0.00
not-MIWAE	0.81 ± 0.00	1.27 ± 0.47	1.19 ± 0.40	1.24 ± 0.35	1.01 ± 0.00
not-MIWAE+ (concat.)	0.76 ± 0.00	1.19 ± 0.43	1.18 ± 0.35	1.20 ± 0.31	1.04 ± 0.00
not-MIWAE+ (additive)	1.20 ± 0.00	1.71 ± 0.51	1.75 ± 0.42	1.61 ± 0.44	0.92 ± 0.00
not-MIWAE+ (multiplicative)	1.84 ± 0.00	1.68 ± 0.16	1.95 ± 0.41	2.43 ± 0.90	0.93 ± 0.00
Mean	1.52 ± 0.00	1.53 ± 0.01	1.51 ± 0.02	1.49 ± 0.05	1.56 ± 0.00
MICE	1.41 ± 0.00	1.50 ± 0.09	1.47 ± 0.09	1.43 ± 0.10	1.29 ± 0.00
missForest	1.24 ± 0.00	1.37 ± 0.13	1.36 ± 0.11	1.35 ± 0.09	1.20 ± 0.00

Table 1: Imputation RMSE (mean $\pm$ std) across UCI datasets affected by conditionally dependent MNAR. Extended not-MIWAE (not-MIWAE+) uses the full encoder. Bold indicates the lowest value for each dataset.

encountered NaN values even after debugging steps such as adding $\epsilon > 0$ to the Gaussian standard deviations.

New correlated MNAR missing mechanism. To test the model’s ability to capture correlated missingness, we introduce a two-level missing mechanism that combines self-censoring with pairwise column dependencies. Specifically, we apply self-censoring to the first half of the features (setting values above their mean to missing), then propagate this missingness to corresponding features in the second half: if feature d ($d < D/2$, where D is the number of columns in the dataset) is missing for observation i , feature $d + D/2$ is also set to missing for i . This process creates explicit dependencies between mask variables that violate the conditional independence assumption (not-MIWAE eqn. 10) of the baseline model. Missing rates for the datasets are shown in Table 2.

Parameters. All models in Table 1 were trained with a hidden layer size of $n_{\text{hidden}} = 128$ and a data latent dimension of $n_{\text{latent}} = D - 1$, which is dataset-dependent. For the extended model, the latent dimension for the regime variable was set to $n_{\text{latent},u} = 10$. Training used $n_{\text{samples}} = 20$ importance samples, a batch size of $\text{batch_size} = 16$, and was run for a maximum of $\text{max_iter} = 100,000$ iterations. For imputation, we drew $L = 5,000$ samples.

Table 2: Missing rates for each dataset.

Dataset	Missing Rate (%)
Banknote	51.35
Concrete	41.88
Red Wine	39.02
White Wine	39.39
Breast	39.86

4.2 Imputation performance with varying training iterations

We initially trained our models with smaller numbers of training iterations, using max_iter values of 1,000 and 10,000, but observed that the extended models did not perform well until the number of iterations was increased to 100,000. Results for 1,000 and 10,000 iterations are reported in the appendix (Tables 4 and 5).

To assess how the number of training iterations affects imputation performance, we select two models, the baseline not-MIWAE and the extended not-MIWAE (concat.), and plot RMSE with

standard deviations as a function of `max_iter`. The results are shown in Figure 3.

We observe a consistent trend across all five datasets. The imputation performance of the extended not-MIWAE is comparable to or substantially worse than that of the baseline not-MIWAE at lower training iterations. However, once the number of iterations exceeds 50,000, the RMSE of the extended not-MIWAE decreases significantly, whereas the baseline not-MIWAE does not exhibit similar improvements. On the contrary, the baseline model’s imputation performance deteriorates for four out of five datasets when `max_iter` is increased from 50,000 to 100,000.

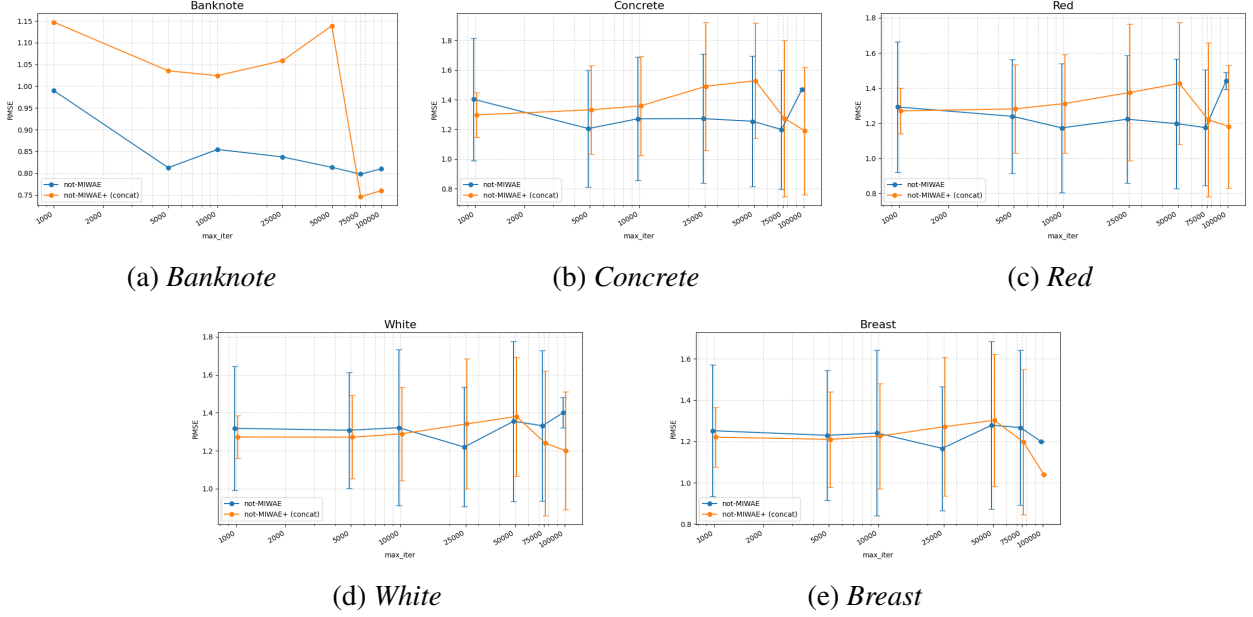


Figure 3: Imputation RMSE of two selected models as a function of `max_iter`.

5 Conclusion

In this project, we extended the not-MIWAE framework to handle correlated missingness by introducing a latent regime variable that captures dependencies between mask variables. Our experiments on multiple UCI datasets demonstrate that the extended model consistently improves imputation performance compared to the baseline not-MIWAE, particularly at higher training iterations. The results highlight the importance of modeling structured missingness in MNAR settings and suggest that incorporating additional latent variables can effectively capture more complex missing patterns. Future work could explore more flexible architectures for the missing predictor and assess performance on larger-scale or real-world datasets.

References

- Burda, Yuri, Roger Grosse, and Ruslan Salakhutdinov. 2015. “Importance weighted autoencoders.” *arXiv preprint arXiv:1509.00519*.
- Ipsen, Niels Bruun, Pierre-Alexandre Mattei, and Jes Frellsen. 2020. “not-MIWAE: Deep generative modelling with missing not at random data.” *arXiv preprint arXiv:2006.12871*.
- Mattei, Pierre-Alexandre, and Jes Frellsen. 2019. “MIWAE: Deep generative modelling and imputation of incomplete data sets.” In *International conference on machine learning*, 4413–4423. PMLR.
- Stekhoven, Daniel J, and Peter Bühlmann. 2012. “MissForest—non-parametric missing value imputation for mixed-type data.” *Bioinformatics* 28 (1): 112–118.
- Van Buuren, Stef, and Karin Groothuis-Oudshoorn. 2011. “mice: Multivariate imputation by chained equations in R.” *Journal of statistical software* 45:1–67.

Appendix

A Mask-only encoder

We conducted an additional experiment in which we trained the models using the same setup and parameter values as in Table 1. The only difference is that we used the mask-only encoder (Equation 8), so that the data x is not used as input when sampling the regime variable u .

Table 3 shows that, even with sufficient training iterations, introducing the regime variable u does not improve imputation performance over not-MIWAE when the mask-only encoder is used.

Model	Banknote	Concrete	Red	White	Breast
MIWAE	1.44 $\pm$ 0.00	1.47 $\pm$ 0.02	1.44 $\pm$ 0.04	1.41 $\pm$ 0.07	1.20 $\pm$ 0.00
not-MIWAE	0.87 $\pm$ 0.00	1.13 $\pm$ 0.26	1.12 $\pm$ 0.22	1.27 $\pm$ 0.33	1.01 $\pm$ 0.00
not-MIWAE+ (concat.)	1.41 $\pm$ 0.00	1.47 $\pm$ 0.06	1.44 $\pm$ 0.06	1.40 $\pm$ 0.08	1.19 $\pm$ 0.00
not-MIWAE+ (additive)	1.09 $\pm$ 0.00	1.72 $\pm$ 0.63	1.94 $\pm$ 0.61	1.75 $\pm$ 0.62	0.87 $\pm$ 0.00
not-MIWAE+ (multiplicative)	1.89 $\pm$ 0.00	1.70 $\pm$ 0.18	1.83 $\pm$ 0.23	1.98 $\pm$ 0.33	0.88 $\pm$ 0.00
Mean	1.52 $\pm$ 0.00	1.53 $\pm$ 0.01	1.51 $\pm$ 0.02	1.49 $\pm$ 0.05	1.56 $\pm$ 0.00
MICE	1.41 $\pm$ 0.00	1.50 $\pm$ 0.09	1.47 $\pm$ 0.09	1.43 $\pm$ 0.10	1.29 $\pm$ 0.00
missForest	1.58 $\pm$ 0.00	1.54 $\pm$ 0.04	1.48 $\pm$ 0.10	1.44 $\pm$ 0.11	1.19 $\pm$ 0.00

Table 3: Imputation RMSE (mean $\pm$ std) across UCI datasets with the mask-only encoder. Bold indicates the lowest value for each dataset.

B Imputation results with lower training iterations (full encoder)

Model	<i>Banknote</i>	<i>Concrete</i>	<i>Red</i>	<i>White</i>	<i>Yeast</i>	<i>Breast</i>
MIWAE	1.43 ± 0.00	1.45 ± 0.02	1.44 ± 0.02	1.41 ± 0.05	1.42 ± 0.05	1.39 ± 0.08
not-MIWAE	0.95 ± 0.00	1.32 ± 0.37	1.25 ± 0.32	1.22 ± 0.28	1.19 ± 0.26	1.15 ± 0.25
not-MIWAE+ (concat)	1.21 ± 0.00	1.33 ± 0.11	1.29 ± 0.11	1.29 ± 0.09	1.31 ± 0.09	1.26 ± 0.13
not-MIWAE+ (add.)	1.07 ± 0.00	1.28 ± 0.21	1.33 ± 0.19	1.34 ± 0.16	1.36 ± 0.15	1.30 ± 0.19
not-MIWAE+ (mult.)	1.67 ± 0.00	1.59 ± 0.08	1.52 ± 0.11	1.48 ± 0.13	1.48 ± 0.11	1.40 ± 0.19
Mean	1.52 ± 0.00	1.53 ± 0.01	1.51 ± 0.02	1.49 ± 0.05	1.48 ± 0.04	1.50 ± 0.05
MICE	1.41 ± 0.00	1.50 ± 0.09	1.47 ± 0.09	1.43 ± 0.10	1.45 ± 0.09	1.42 ± 0.10
missForest	1.31 ± 0.00	1.41 ± 0.09	1.38 ± 0.08	1.36 ± 0.08	1.39 ± 0.09	1.36 ± 0.11

Table 4: Imputation RMSE (mean $\pm$ std) across UCI datasets. `max_iter` = 1,000.

Model	<i>Banknote</i>	<i>Concrete</i>	<i>Red</i>	<i>White</i>	<i>Yeast</i>	<i>Breast</i>
MIWAE	1.41 ± 0.00	1.46 ± 0.05	1.43 ± 0.05	1.40 ± 0.07	1.41 ± 0.07	1.38 ± 0.11
not-MIWAE	0.86 ± 0.00	1.45 ± 0.59	1.31 ± 0.52	1.52 ± 0.59	1.43 ± 0.56	1.34 ± 0.55
not-MIWAE+ (concat)	1.06 ± 0.00	1.45 ± 0.39	1.35 ± 0.35	1.33 ± 0.31	1.35 ± 0.28	1.29 ± 0.29
not-MIWAE+ (add.)	1.41 ± 0.00	1.74 ± 0.33	1.56 ± 0.37	1.54 ± 0.32	1.53 ± 0.29	1.43 ± 0.34
not-MIWAE+ (mult.)	1.85 ± 0.00	1.74 ± 0.11	1.72 ± 0.10	1.63 ± 0.18	1.61 ± 0.17	1.49 ± 0.30
Mean	1.52 ± 0.00	1.53 ± 0.01	1.51 ± 0.02	1.49 ± 0.05	1.48 ± 0.04	1.50 ± 0.05
MICE	1.41 ± 0.00	1.50 ± 0.09	1.47 ± 0.09	1.43 ± 0.10	1.45 ± 0.09	1.42 ± 0.10
missForest	1.46 ± 0.00	1.49 ± 0.03	1.44 ± 0.07	1.41 ± 0.08	1.44 ± 0.08	1.40 ± 0.12

Table 5: Imputation RMSE (mean $\pm$ std) across UCI datasets. `max_iter` = 10,000.